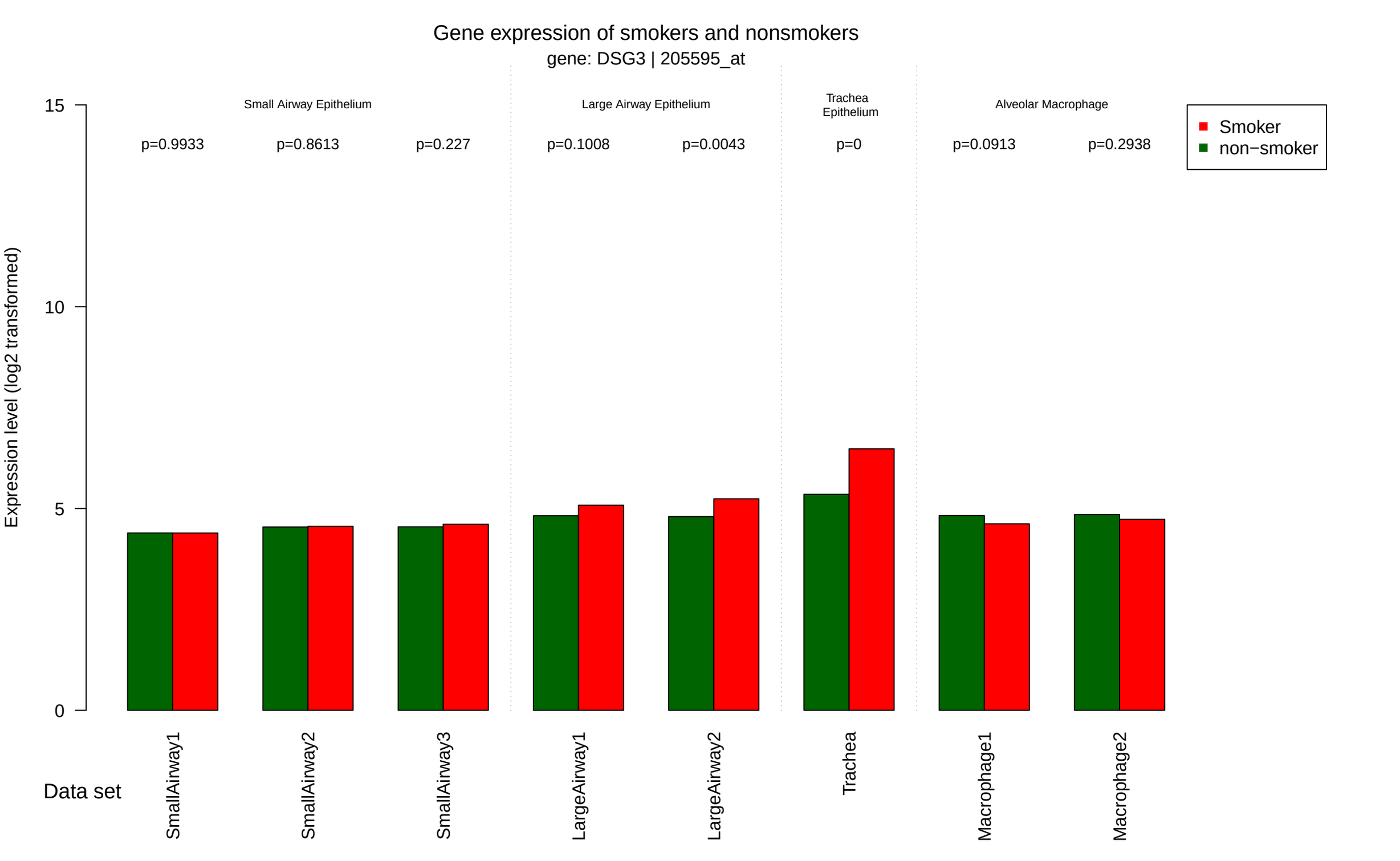
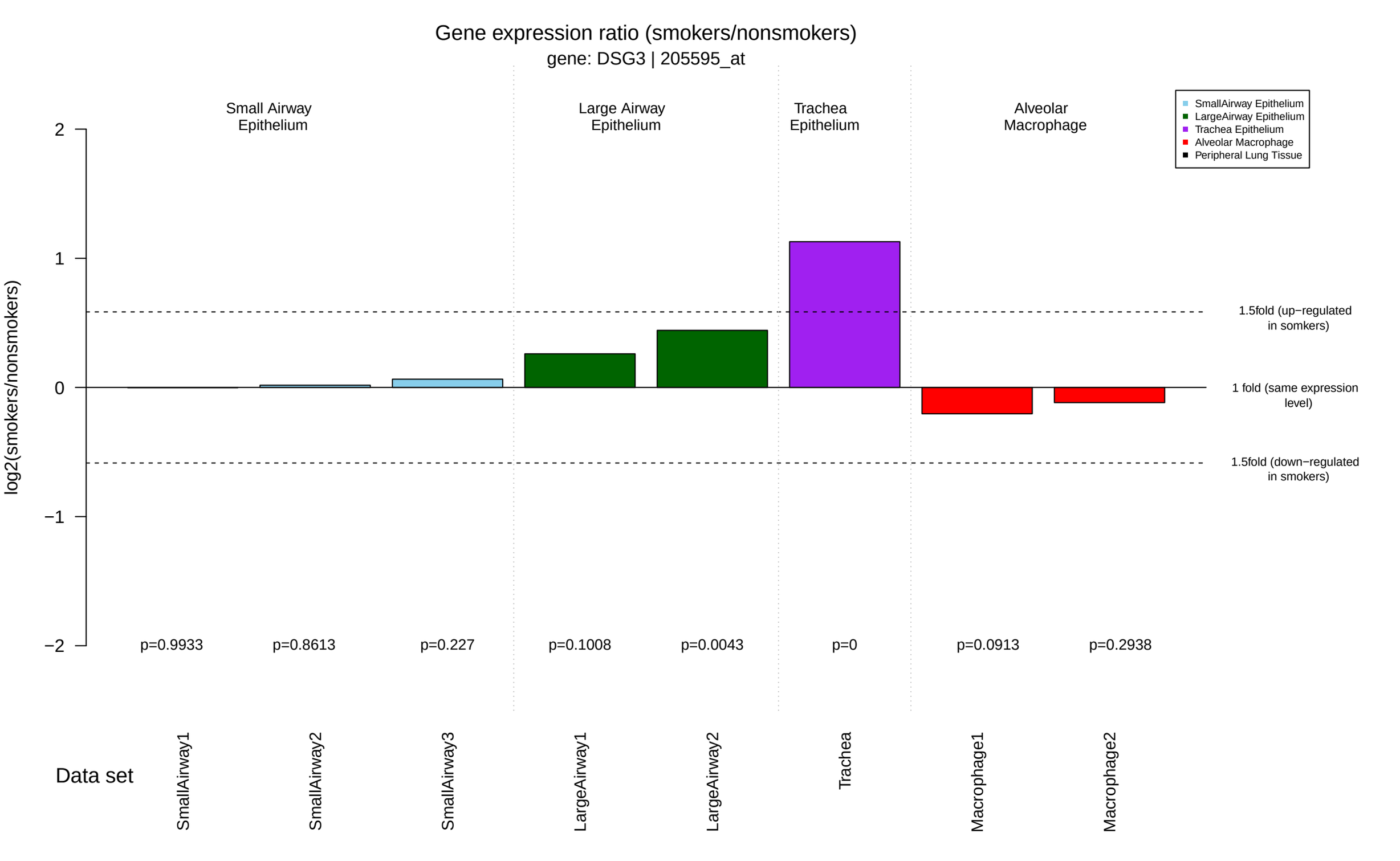
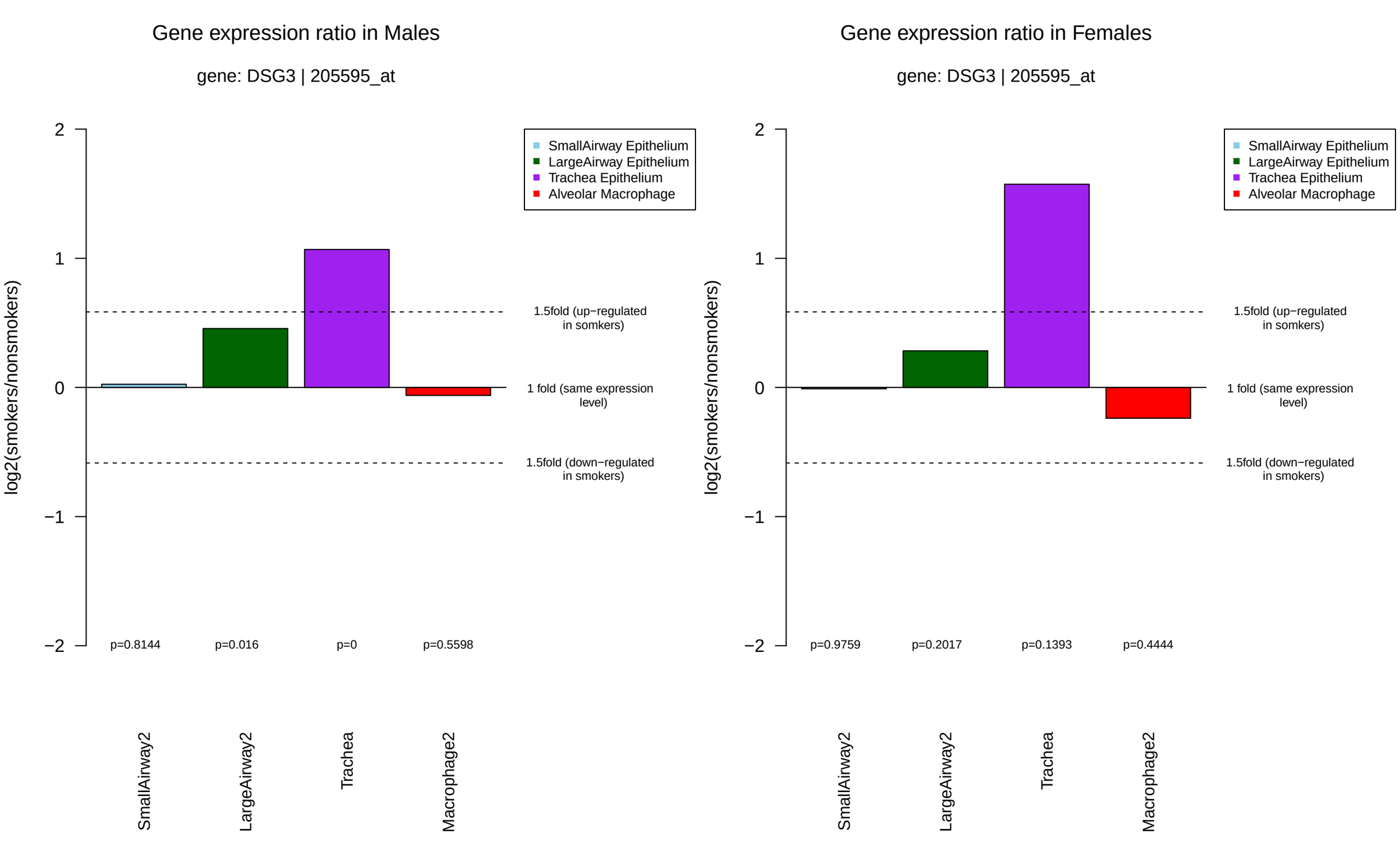
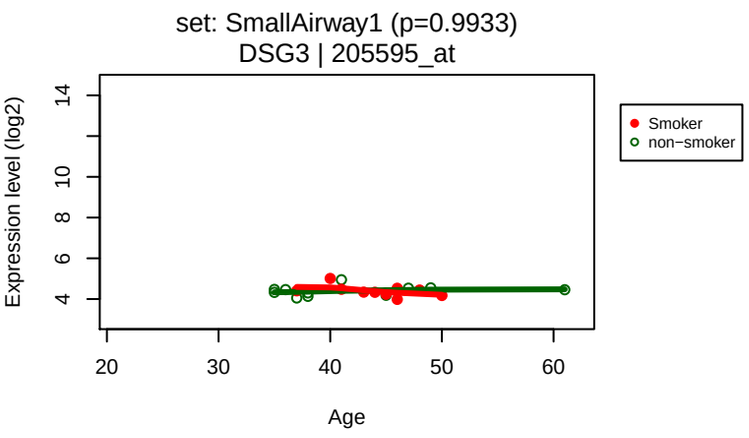
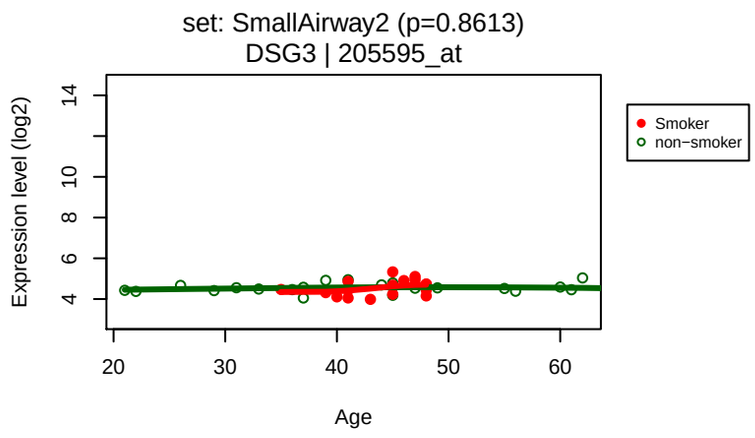
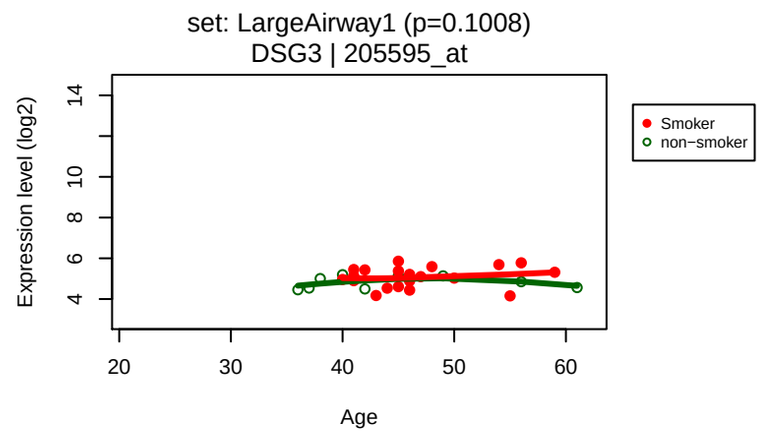

Gene expression of smokers and nonsmokers
gene: DSG3 | 205595_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9933
p=0.8613
p=0.227
p=0.1008
p=0.0043
p=0
p=0.0913
p=0.2938
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

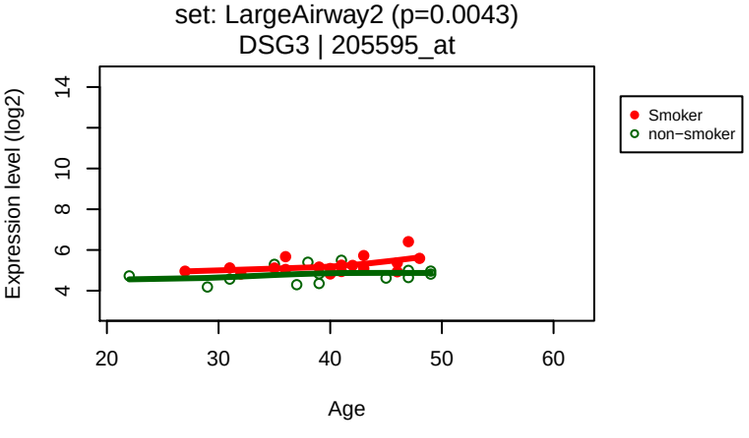

Gene expression ratio (smokers/nonsmokers)
gene: DSG3 | 205595_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9933
p=0.8613
p=0.227
p=0.1008
p=0.0043
p=0
p=0.0913
p=0.2938
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

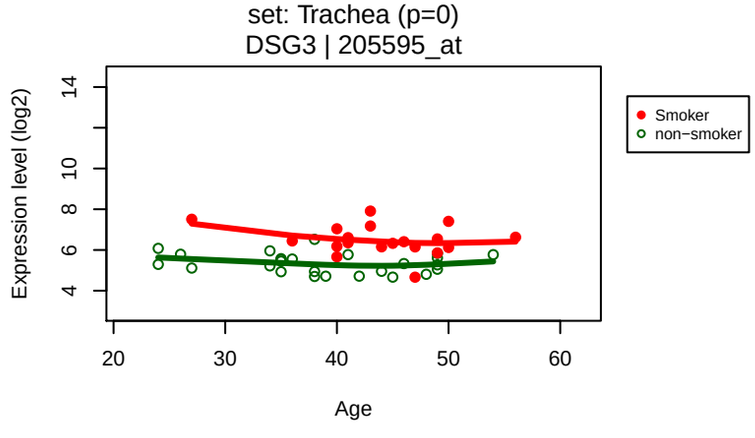

Gene expression ratio in Males
gene: DSG3 | 205595_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8144
p=0.016
p=0
p=0.5598
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DSG3 | 205595_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9759
p=0.2017
p=0.1393
p=0.4444
SmallAirway2
LargeAirway2
Trachea
Macrophage2

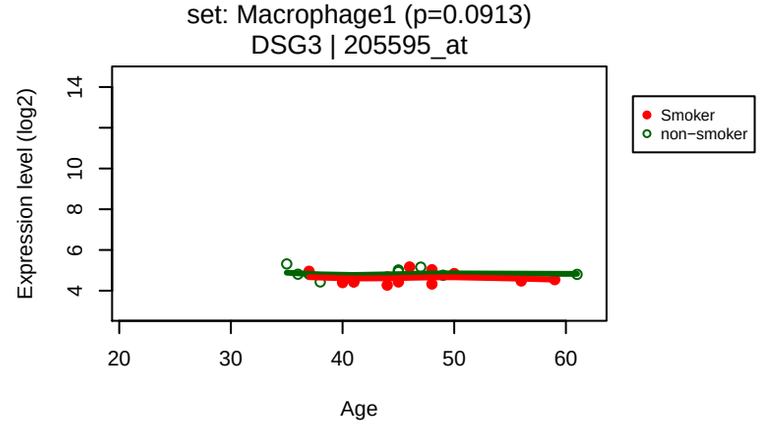

set: SmallAirway1 (p=0.9933)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker

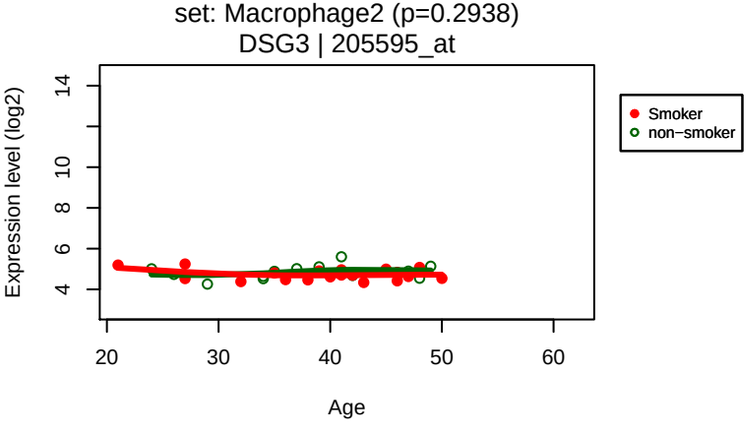

set: SmallAirway2 (p=0.8613)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker

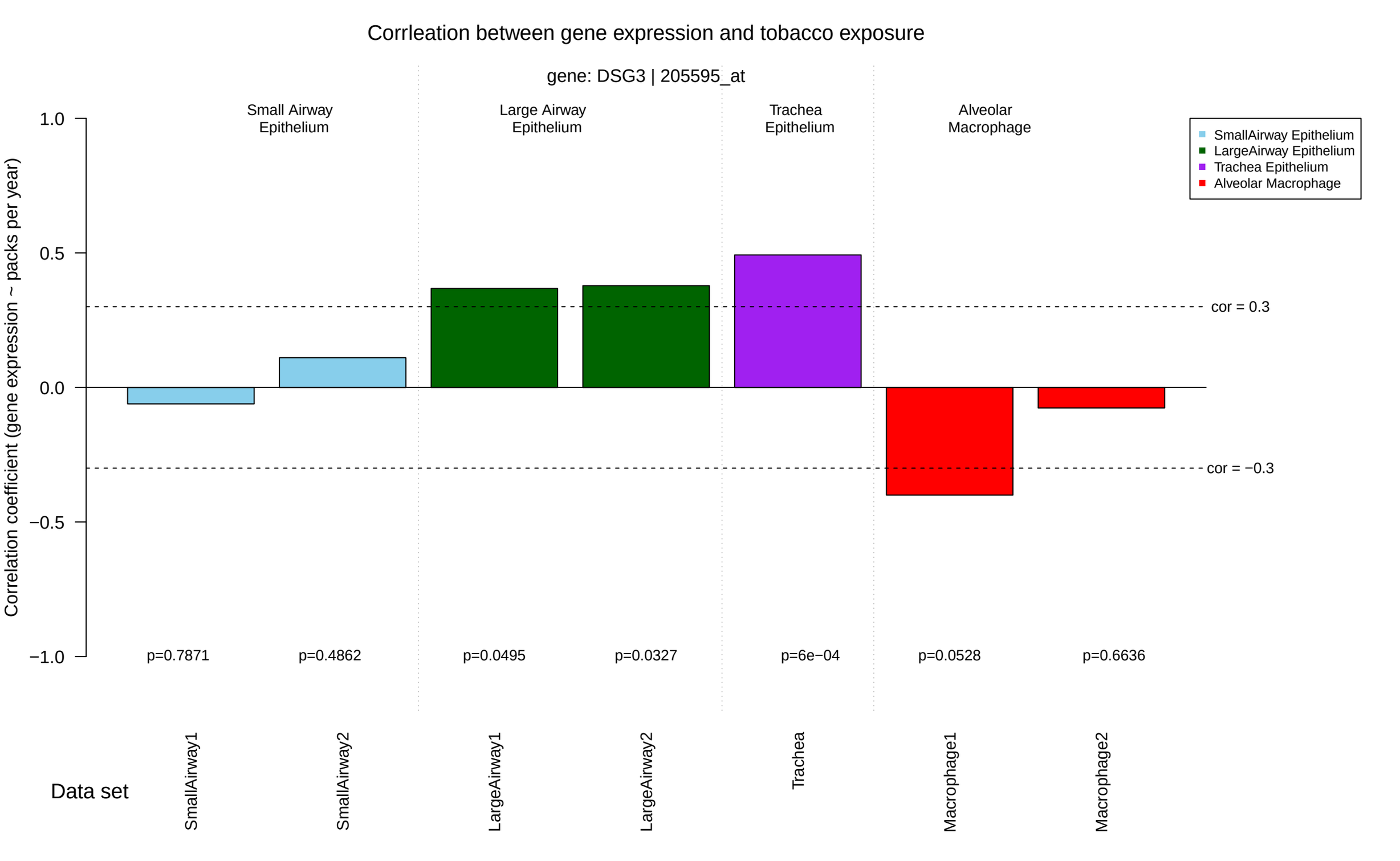

set: LargeAirway1 (p=0.1008)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0043)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0913)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2938)
DSG3 | 205595_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DSG3 | 205595_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7871
p=0.4862
p=0.0495
p=0.0327
p=6e−04
p=0.0528
p=0.6636
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2